

Review

Plasma-Assisted Hydrogen Production: Technologies, Challenges, and Future Prospects

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Abstract: As global demand for clean energy continues to rise, hydrogen, as an ideal energy carrier, plays a crucial role in the energy transition. Traditional hydrogen production methods predominantly rely on fossil fuels, leading to environmental pollution and energy inefficiency. In contrast, plasma-assisted hydrogen production, as an emerging technology, has gained significant attention due to its high efficiency, environmental friendliness, and flexibility. Plasma technology generates high-energy electrons or ions by exciting gas molecules, which, under specific conditions, effectively decompose water vapor or hydrocarbon gases to produce hydrogen. This review systematically summarizes the basic principles, technological routes, research progress, and potential applications of plasma-assisted hydrogen production. It focuses on various plasma-based hydrogen production methods, such as water vapor decomposition, hydrocarbon cracking, arc discharge, and microwave discharge, highlighting their advantages and challenges. Additionally, it addresses key issues facing plasma-assisted hydrogen production, including energy efficiency improvement, reactor stability, and cost optimization, and discusses the future prospects of these technologies. With ongoing advancements, plasma-assisted hydrogen production is expected to become a mainstream technology for hydrogen production, contributing to global goals of zero carbon emissions and sustainable energy development.

Keywords: plasma hydrogen production; water decomposition; hydrocarbon cracking; plasma technologies



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1. Introduction

1.1. The Potential of Plasma Technology in Hydrogen Production

Hydrogen, as a clean and efficient energy carrier, plays a pivotal role in the transition to a sustainable energy future. Traditional hydrogen production methods, such as steam reforming and coal gasification, are widely used due to their cost-effectiveness and scalability. However, these methods are associated with significant carbon emissions and energy inefficiencies, making them less sustainable in the long run. In contrast, plasma-assisted hydrogen production technologies have emerged as a promising alternative. These technologies use high-energy electrons and ions to decompose water vapor or hydrocarbons, offering greater efficiency and lower environmental impact. Unlike conventional methods, plasma-assisted technologies operate at lower temperatures and pressures, making them more suitable for integration with renewable energy sources.

1.2. The Definition and Fundamental Properties of Plasma

Plasma, the fourth state of matter, differs from solid, liquid, and gas phases, as illustrated in Figure 1. It consists of free electrons, ions, and neutral particles, exhibiting distinct electromagnetic properties. Plasma typically forms under high-temperature or low-pressure conditions, where atoms or molecules in a gas acquire enough energy to eject electrons, resulting in positively charged ions and free electrons. Due to the largely charged nature of the particles in plasma, it is responsive to external electromagnetic fields, exhibiting a range of physical characteristics that differ significantly from those of gases. Plasma is abundant in the universe, found in interstellar space, the Sun, and the core of other stars, while on Earth, plasma can be generated in laboratory settings through processes such as arc discharge or flames. Plasma has important applications in fields such as nuclear fusion, semiconductor manufacturing, and medicine [1]. Research has shown that plasma technology can significantly enhance the degradation efficiency of ibuprofen, thereby providing new insights into pollutant treatment in various fields [2]. Additionally, in-liquid plasma is expected to be a promising method for enhancing surface and electrochemical properties [3].

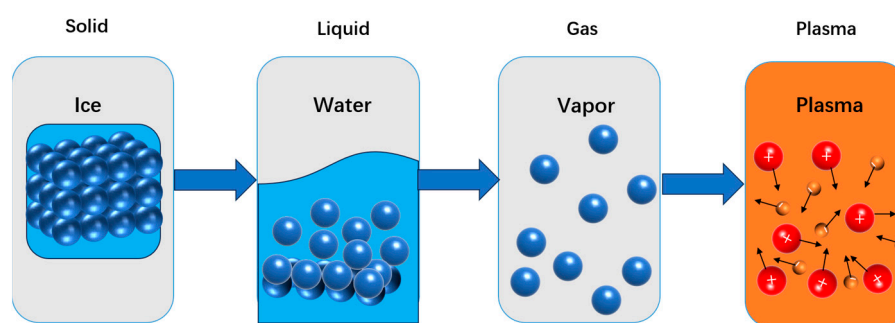


Figure 1. Schematic diagram of the four states of matter.

Plasma can be classified from various perspectives based on its distinct characteristics and applications. Common classification methods include categorizing plasma according to temperature, density, degree of ionization, and its interaction with external magnetic fields. The following are several primary classification approaches for plasma.

Owing to these characteristics, plasma holds significant potential for applications in hydrogen production of free electrons and ions, and demonstrates a range of distinctive physical properties. The following section outlines the fundamental characteristics of plasma pertinent to hydrogen production, along with detailed explanations (as illustrated in Figure 2).

(1) High energy density and excitation properties. Plasma electrons possess elevated energy levels, which enable them to excite and decompose gas molecules, such as water molecules. High-energy electrons in plasma collide with gas molecules, exciting them to high-energy states or ionizing them, thereby inducing molecular dissociation or the generation of free radicals. This property is one of the fundamental principles of plasma-assisted hydrogen production, where water molecules (H_2O) can be decomposed into hydrogen and oxygen through excitation. The high-energy electrons in plasma disrupt the O-H chemical bonds in water, leading to the decomposition of the molecules into hydrogen atoms (H) and hydroxide ions (OH^-). These free radicals subsequently recombine during the reaction to form hydrogen gas [4].

(2) High electrical conductivity and electric field responsiveness. The electrical conductivity of plasma is attributed to the presence of free electrons and ions, which allows plasma to respond to external electric fields. When an external electric field is applied to plasma, electrons and ions accelerate under the field's influence, supplying sufficient

energy for reactions within the plasma. Furthermore, fluctuations in the electric field within the plasma can influence the collision frequency of gas molecules, thereby increasing the reaction efficiency. Within plasma, the electric field accelerates the motion of electrons and ions and also enhances the frequency of collisions between ions and molecules, thereby facilitating hydrogen production [5]. For instance, applying alternating electric fields to excite plasma can decompose water molecules at reduced temperatures, significantly lowering the high-temperature requirements of traditional chemical methods.



Figure 2. Fundamental properties of plasma in hydrogen production.

(3) Non-thermal equilibrium state. Plasma contains charged particles (such as electrons and ions) and high-energy free radicals (e.g., hydrogen atoms and hydroxide ions), both of which exhibit exceptionally high reactivity. The intense ionization effects of plasma enable vigorous reactions among molecules, atoms, and free radicals in the gas phase, rendering plasma an ideal medium for promoting hydrogen production. Plasma generates a substantial number of free radicals through ionization, which exhibit intense chemical reactivity and can rapidly interact with water molecules to yield hydrogen gas. Notably, hydrogen atoms, as highly reactive free radicals, can be further combined with other hydrogen molecules to produce hydrogen gas [6].

(4) Reactivity and chemical reactivity. Plasma contains highly reactive charged particles, including electrons, ions, as well as high-energy free radicals like hydrogen atoms and hydroxyl ions. The strong ionization effect of plasma induces intense reactions among molecules, atoms, and free radicals in the gas phase, establishing plasma as an ideal medium for hydrogen production [7]. Plasma, through ionization, generates abundant free radicals, which possess high chemical reactivity and can rapidly interact with water molecules to generate hydrogen. Specifically, hydrogen atoms, being highly reactive radicals, can further combine with other hydrogen molecules to form hydrogen gas.

(5) Controllability and flexibility. The reaction conditions in plasma can be finely tuned by manipulating external factors, such as electric fields, power density, and other parameters, thereby enabling precise control over the hydrogen production rate and product distribution. This flexibility renders plasma an exceptionally ideal technology for hydrogen production. By adjusting factors such as input power, gas composition, and pressure, the plasma reaction process can be optimized [8].

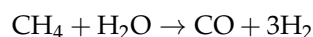
(6) The high reactivity exhibited under low-pressure conditions. Low-pressure plasmas are typically characterized by higher reactivity, allowing for efficient hydrogen generation with reduced energy consumption. Under low-pressure conditions, plasma facilitates a more uniform energy distribution, which enhances the dissociation of gas molecules more efficiently. In microwave plasmas, a low-pressure environment can substantially enhance the efficiency of water decomposition, resulting in a marked increase in hydrogen yield [9].

1.3. Comparison of Traditional Hydrogen Production and Plasma-Assisted Hydrogen Production

1.3.1. Fundamental Principles of Traditional Hydrogen Production

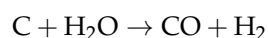
In response to the environmental challenges posed by conventional hydrogen production methods, plasma-assisted hydrogen production technologies have emerged as a promising alternative. These technologies harness high-energy electrons and ions to decompose water vapor or hydrocarbons, offering greater efficiency, lower environmental impact, and potential integration with renewable energy sources. Plasma-assisted technologies operate at lower temperatures and pressures compared to traditional methods, significantly reducing energy consumption and improving the overall sustainability of hydrogen production. Unlike traditional methods, which rely on high-temperature processes, plasma systems can generate hydrogen more efficiently, especially when coupled with renewable energy, thus advancing the goal of zero-carbon hydrogen production.

Steam reforming and coal gasification are the primary methods used for traditional hydrogen production. Steam reforming is currently the most widely employed commercial hydrogen production method, which involves reacting natural gas (primarily methane) with steam to generate hydrogen. The reaction equation is as follows:



In this process, methane reacts with steam under the influence of a catalyst, generating hydrogen and carbon monoxide. This method offers several advantages, including technological maturity, widespread use of large-scale equipment, and low costs. However, it also produces significant carbon dioxide emissions, contributing to environmental pollution [10].

Coal gasification involves reacting coal with oxygen or steam at high temperatures to produce syngas, primarily consisting of hydrogen, carbon monoxide, and carbon dioxide. The reaction equation is as follows:



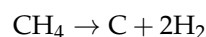
Although coal gasification can efficiently produce hydrogen, its reliance on fossil fuels results in considerable environmental impacts, along with high costs and significant energy consumption [11].

1.3.2. Fundamental Principles of Plasma-Assisted Hydrogen Production

Plasma-assisted hydrogen production involves high-energy particles in plasma (such as high-energy electrons, ions, and free radicals) reacting with hydrogen sources (such as water, methane, etc.) to generate hydrogen gas. The process typically depends on high-temperature electrons in plasma colliding with gas molecules, breaking the chemical bonds within the molecules to generate hydrogen gas and other by-products. Plasma hydrogen production is typically achieved through methods such as microwave plasma or direct current discharge. The following introduces two methods to provide a general overview of their fundamental principles.

One method is water vapor decomposition, where high-energy electrons in the plasma excite water molecules, causing ionization and generating hydrogen atoms and hydroxyl ions, ultimately forming hydrogen and oxygen gas. This process typically occurs at lower temperatures, making it energy-efficient and environmentally friendly [12].

Another method is methane cracking, where plasma can break down methane to generate hydrogen gas and solid carbon, thus avoiding the carbon dioxide emissions typical of traditional methods. The reaction equation is as follows:



This method results in a lower environmental impact during hydrogen production, and the carbon produced can be utilized in other applications [13].

1.3.3. Additional Comparisons

Traditional commercial hydrogen production methods, such as steam reforming and coal gasification, dominate the industry; however, these methods generally exhibit high energy consumption and carbon emissions. Plasma-assisted hydrogen production, by contrast, utilizes high-energy electron plasmas to achieve efficient hydrogen production at lower temperatures and pressures, resulting in lower carbon emissions and enhanced energy efficiency. Furthermore, while the equipment cost of plasma-assisted hydrogen production is higher and still faces certain technological hurdles, its environmental and energy efficiency advantages position it as a formidable competitor for future hydrogen production technologies [14]. As technology progresses, plasma-assisted hydrogen production is expected to become a more widely adopted method for hydrogen production.

1.3.4. Innovation and Research Contributions

This review distinguishes itself from existing literature by systematically analyzing the latest advancements in plasma-assisted hydrogen production, focusing on the integration of plasma technologies with renewable energy systems, such as water electrolysis. Unlike many existing studies that primarily focus on the basic principles or isolated applications of plasma technologies, this review provides a comprehensive discussion on the challenges faced in terms of energy efficiency, reactor stability, and cost optimization. Furthermore, it addresses the gap in current research by exploring the potential of plasma-assisted hydrogen production in combination with renewable energy sources, presenting new insights into the integration of these technologies for a more sustainable and low-carbon hydrogen production future.

Through this review, we aim to provide a clearer understanding of the current state of plasma-assisted hydrogen production, highlight its technological advancements, and identify the key research gaps that need to be addressed to advance this field. These include improving the energy efficiency of plasma systems, enhancing reactor stability, and optimizing operational costs, which are critical to making plasma-based hydrogen production a commercially viable and environmentally sustainable technology.

2. Plasma-Assisted Hydrogen Production Methods

Plasma is a high-energy state of matter composed of charged particles, free electrons, and neutral atoms or molecules. At high temperatures, plasma can provide sufficient energy to break the chemical bonds of water molecules or hydrocarbon molecules, thereby producing hydrogen. The core of plasma-based hydrogen production technology lies in exciting gas molecules through external electric or magnetic fields, generating high-energy ions and free radicals, which then undergo decomposition reactions under appropriate conditions.

2.1. Plasma-Assisted Steam Cracking

Plasma-assisted steam cracking for hydrogen production is an advanced method that utilizes the high-energy excitation of plasma to break down steam molecules, generating hydrogen. This process typically occurs in a high-temperature, high-energy plasma environment, where steam interacts with reactive species in the plasma, such as electrons, ions, and free radicals. This interaction leads to the dissociation of water (H_2O) into hydrogen (H_2) and oxygen (O_2) [15]. Four common processes that lead to water decomposition near the discharge electrode are given in Figure 3. Compared to traditional thermal cracking and water electrolysis methods, plasma-assisted steam cracking offers lower energy consumption and higher reaction efficiency, making it a promising hydrogen production technology.

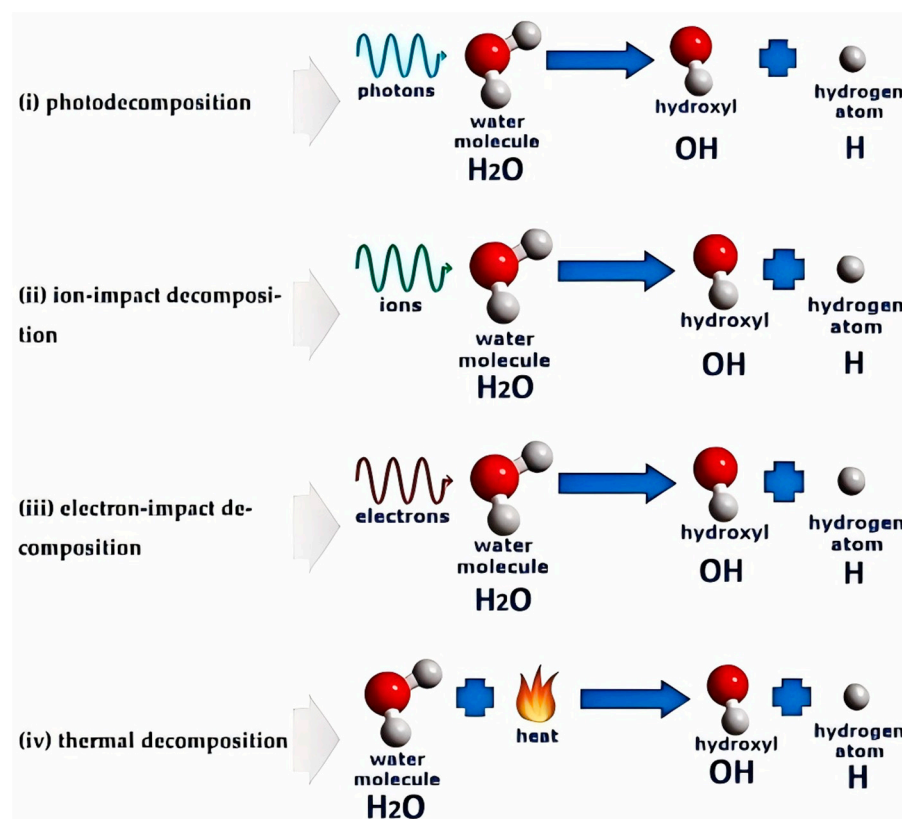


Figure 3. Illustration of the four processes leading to the water decomposition in the vicinity of the discharge electrode [16].

The core of plasma-assisted steam cracking for hydrogen production lies in the generation and control of plasma. Common plasma sources include direct current (DC) discharge, radio frequency (RF) discharge, microwave plasma, and arc discharges. Under the influence of plasma, the O-H bonds in steam molecules are broken, leading to the generation of a large number of hydrogen atoms (H), hydrogen molecules (H_2), oxygen atoms (O), and oxygen molecules (O_2). Hydrogen production not only depends on the temperature, flow rate, and pressure conditions of the steam but is also influenced by plasma parameters, such as power, frequency, and electron density [17].

The advantage of plasma-assisted steam cracking for hydrogen production lies in its highly efficient reaction mechanism. The high temperature (reaching several thousand degrees Celsius) and high energy of plasma enable the dissociation of water molecules with lower energy consumption, while the reaction rate is much higher than that of conventional heating methods [18]. Additionally, by optimizing the design and control of the plasma source, hydrogen yield can be significantly improved, and the formation of by-products such as CO or CO_2 can be minimized. Therefore, plasma-assisted steam cracking is not

only an efficient hydrogen production technology but also holds promise as a supplement to renewable energy, particularly when combined with water electrolysis in solar or wind-driven power systems, offering excellent application potential.

However, plasma-assisted steam cracking for hydrogen production still faces several challenges, including the stability of the plasma system, the further improvement in energy efficiency, and the enhancement of hydrogen yield. To address these issues, researchers are exploring different types of plasma sources [19,20], reactor designs [21], and optimization of operational parameters [22] to achieve more efficient hydrogen production [21]. Additionally, the development of efficient power systems and suitable catalysts can also contribute to improving the overall performance and economics of this technology.

Overall, plasma-assisted steam cracking for hydrogen production, as an advanced technology, demonstrates significant potential for future hydrogen production due to its lower energy consumption and higher hydrogen production efficiency. With continuous technological advancements, plasma-assisted steam cracking is expected to play a crucial role in the hydrogen energy industry and contribute to the development of sustainable energy.

2.2. Plasma-Assisted Hydrocarbon Cracking

Plasma-assisted hydrocarbon cracking is a process that uses plasma technology to decompose hydrocarbons (such as methane, ethane, propane, etc.) into simpler molecules or atoms, and subsequently generate hydrogen through corresponding chemical reactions. Figure 4 shows the process of hydrogen and aromatics recovery by plasma catalytic cracking of waste polypropylene. Given the important role of hydrogen as a clean energy source in energy conversion and storage [23], plasma-assisted hydrocarbon cracking has garnered significant attention in the field of hydrogen production.

Carbon-hydrocarbon cracking essentially involves the cleavage of C-C bonds to decompose long-chain hydrocarbons, including both thermal cracking (free radical mechanism) and catalytic cracking (acid-base catalysis mechanism) [24]. Catalytic cracking significantly reduces the activation energy of the reaction (by about 30–50%) and increases the yield of target products at lower temperatures (450–550 °C). The key catalytic mechanisms can be classified into acidic site effects, pore confinement effects, and metal–acid synergism [25].

Plasma cracking technology ionizes gases and generates high-energy electrons, ions, free radicals, and excited molecules. These reactive species efficiently break the chemical bonds of hydrocarbon gases at low temperatures, leading to the production of hydrogen, olefins, and other by-products. Compared to conventional thermal cracking methods, the main advantage of plasma cracking lies in its ability to achieve effective hydrocarbon gas dissociation under low-temperature conditions, while maintaining low energy consumption and high reaction rates [26].

The acidic site effect mainly refers to the role of Brønsted acid sites in zeolite catalysts (such as H-ZSM-5, Y-type zeolite), which initiate β -scission by protonating the hydrocarbon chain, resulting in small-molecule olefins [27]. Y-type zeolites, due to their high acidity and large pore size, are widely used in fluidized catalytic cracking (FCC), but they face issues related to micropore diffusion limitations and coke deactivation. The pore confinement effect refers to the role of the microporous structure (0.5–1.2 nm) of molecular sieves, which promotes the adsorption and reaction of specific-sized molecules through spatial restriction [28], enhancing product selectivity (e.g., MCM-41, SBA-15). By introducing mesopores (2–50 nm), the diffusion efficiency of large molecules is improved, making them suitable for heavy oil cracking, with a conversion rate improvement of 15–30%. Metal–acid synergism: Catalysts loaded with metals (such as Pt, Mo) can adjust product distribution through dehydrogenation/hydrogen transfer reactions [29], suppressing coke

formation. For example, zeolite/metal oxide ($\text{SiO}_2\text{-Al}_2\text{O}_3$) composite systems, combining acidity and thermal stability, extend catalyst life. Nano-scale CeO_2 or TiO_2 enhances the oxidation-reduction ability by providing a high surface area and oxygen vacancies, reducing coke formation. The integration of catalysts into hydrocarbon cracking has evolved from single acidic regulation to multi-scale cooperative design, significantly improving reaction efficiency through pore engineering, metal-acid synergy, and anti-coke strategies.

The plasma cracking process of hydrocarbon gases generally involves three main steps: First, high voltage or high-frequency current ionizes the hydrocarbon gas in the electric field, generating electrons, ions, and high-energy neutral particles. Then, these high-energy particles collide with gas molecules, causing the carbon-hydrogen bonds in the hydrocarbon molecules to break, producing hydrogen, methane, and other short-chain hydrocarbon molecules. Finally, through a series of chemical reactions, the generated products may undergo further decomposition into smaller molecules or stable substances.

Currently, research on plasma cracking of hydrocarbon gases primarily focuses on two major directions. First, improving the selective yield of hydrogen, and second, optimizing energy efficiency. To enhance the selective yield of hydrogen, researchers have explored various plasma sources [30,31] (such as pulsed spark plasma, microwave plasma, etc.) and different atmospheric control methods (such as the use of inert gas assistance). Additionally, some studies have investigated the addition of catalysts to improve cracking efficiency and selectivity [32], further enhancing the reactivity of hydrocarbon gas cracking.

Despite the significant advantages of plasma cracking hydrocarbon gases for hydrogen production, this technology still faces several challenges. These include suboptimal energy consumption ratios, process instability, and high equipment costs. Although arc plasmas excel in terms of pure energy consumption and gas production efficiency, their high operational energy demands may render them economically unfavorable for long-term use [33]. Microwave [34], radio frequency (RF) [35,36], and dielectric barrier discharge (DBD) plasmas [23] offer superior energy efficiency; however, their gas production rates are relatively low. Consequently, achieving an optimal balance between hydrogen production rate and energy consumption remains a critical issue in current research and industrial applications. For instance, hybrid plasma technologies, thermoelectric systems, the integration of nanomaterials and catalysts, as well as innovative reactor designs, could potentially enhance both the efficiency and economic viability of hydrogen production.

Although plasma cracking of hydrocarbon gases offers significant advantages in hydrogen production, the technology still faces several challenges, including high energy consumption, poor process stability, and high equipment costs. Future research will continue to focus on gaining a deeper understanding of the reaction mechanisms, optimizing plasma sources, and improving reaction conditions, with the goal of achieving more efficient and economical hydrogen production.

In conclusion, plasma cracking of hydrocarbon gases presents a promising potential pathway for hydrogen production, showing good prospects. However, further optimization and breakthroughs are still required in terms of energy efficiency and economic viability.

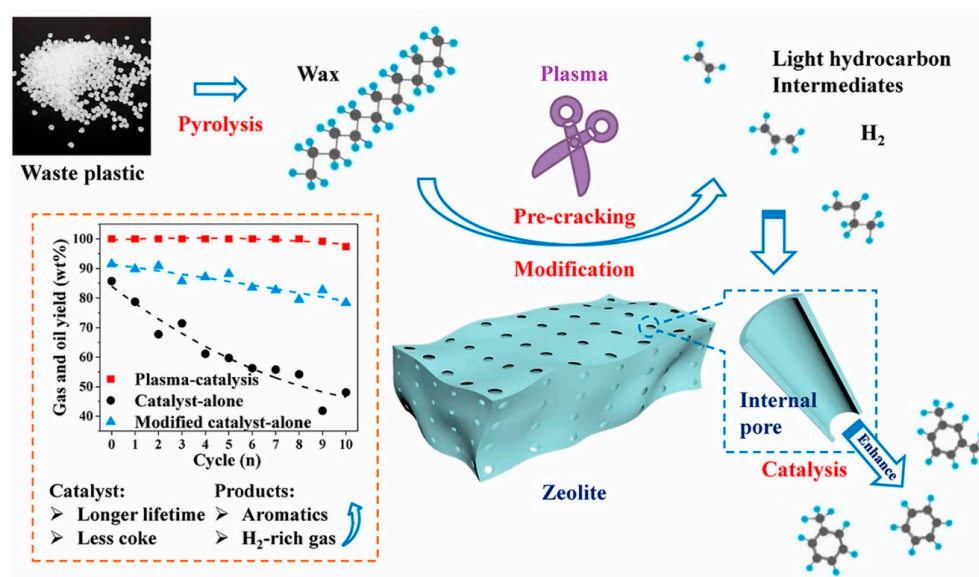


Figure 4. Plasma-catalyzed cracking of waste polypropylene for hydrogen and aromatic hydrocarbons recovery [37].

3. Plasma Generation Techniques for Hydrogen Production

3.1. Arc Discharge

Arc discharge is a technique that generates plasma in a gas through high-voltage electrical arcs, which are used for steam reforming or hydrocarbon gas cracking to produce hydrogen. This method utilizes high-energy electrons and high-temperature environments to excite gas molecules, thereby efficiently decomposing target compounds and generating hydrogen and other by-products.

Arc discharge can be classified into two main types: sliding arc discharge and alternating current (AC) arc discharge. Sliding arc discharge can generate non-equilibrium plasma under ambient temperature and pressure, providing high-energy electrons and reactive free radicals, which significantly increase the reaction rate and efficiency [38,39]. AC arc discharge, on the other hand, achieves higher electron density and average electron energy through localized intense discharge, enabling it to directly break the chemical bonds of target molecules, such as the N-H bonds in ammonia, thereby enhancing the decomposition efficiency [40].

One key advantage of arc discharge technology is its ability to break through the thermodynamic equilibrium limitations and achieve efficient hydrogen production under low or ambient temperature conditions. Figure 5a–c show the flat reactor structure: the flat reactor adopts a 3 mm thick stainless steel plate (50 mm diameter) as the high voltage pole and quartz flakes (50 mm diameter, 2 mm thick) as the dielectric; the grounding electrode selects a flat plate/spiked rod/steel pipe (3 mm diameter) to form a flat–flat, point–flat or tube–flat structure. When the high-voltage pole completely covers the dielectric to produce uniform dielectric barrier discharge, not covered or center openings are generated when the AC arc discharge. The tube–tube reactor (Figure 5d) uses dual stainless steel tube electrodes to produce only AC arc discharges. The height of the gas-passing portion within the reaction zone (flat type) or the spacing of the tube openings (tube–tube type) is the gap distance. This characteristic allows it to exhibit higher energy utilization efficiency compared to traditional catalytic methods. Research by Yan et al. [41] has shown that sliding arc discharge can nearly completely decompose ammonia gas, significantly reducing energy consumption. Furthermore, when combined with catalysts, arc discharge reactors demon-

strate even higher energy efficiency. For example, non-thermal arc discharge combined with NiO/Al₂O₃ catalysts achieves a hydrogen energy efficiency of up to 1080 L/kWh [42].

The discharge mode and reactor design are crucial to the discharge performance. Zhao et al. [40] discussed the variation in ammonia decomposition conversion rates under different discharge modes and reactor structures (Figure 5). Figure 5a–c show the flat reactor structure: the flat reactor adopts a 3 mm thick stainless steel plate (50 mm diameter) as the high voltage pole and quartz flakes (50 mm diameter, 2mm thick) as the dielectric; the grounding electrode selects a flat plate/spiked rod/steel pipe (3mm diameter) to form a flat–flat, point–flat, or tube–flat structure. When the high-voltage pole completely covers the dielectric to produce uniform dielectric-barrier discharge, not covered or center openings are generated when the AC arc discharge. The tube–tube reactor (Figure 5d) uses dual stainless steel tube electrodes to produce only AC arc discharges. The height of the gas-passing portion within the reaction zone (flat type) or the spacing of the tube openings (tube–tube type) is the gap distance. The study found that the discharge mode is a key factor influencing the ammonia decomposition conversion rate. In a plate-type reactor, the ammonia conversion rate under AC arc discharge mode was significantly higher than that under dielectric barrier discharge mode. Additionally, the reactor structure also impacts the ammonia decomposition efficiency, particularly under AC arc discharge mode, where the use of a perforated dielectric further improves the ammonia conversion rate. Under AC arc discharge mode, the ranking of reactor structures in terms of ammonia decomposition efficiency is as follows: tube–tube > tube–plate > needle–plate > plate–plate.

Although the arc discharge method demonstrates excellent performance in improving reaction efficiency and reducing energy consumption, it still faces certain limitations and challenges. First, high power input may lead to significant energy losses, such as thermal energy dissipation, which could increase operational costs in industrial-scale applications [42]. Second, the selection of electrode materials and reactor design requires further optimization to enhance long-term stability and reduce electrode corrosion [40]. Regarding the specific optimization paths for arc discharge technology, there is some debate in the literature. Some scholars suggest that improving electron density and local energy density is key, while others focus on optimizing reactor materials and structures. Furthermore, the relationship between discharge conditions and reaction pathways requires further validation through theoretical models [38,39].

Through continuous research and improvements, the arc discharge method is expected to become an efficient and low-carbon hydrogen production technology, especially as it exhibits significant advantages in processing complex compounds or low-quality fuels. Future development trends should focus on the following aspects: (1) improving energy efficiency by integrating heterogeneous catalysts and operating under negative pressure; (2) optimizing reactor design to maximize the uniformity and stability of the discharge zone; (3) developing low-cost and high-efficiency arc discharge systems suitable for large-scale industrial applications.

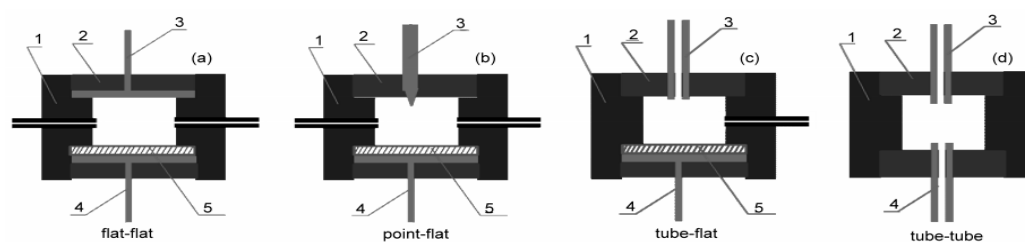


Figure 5. Scheme of configuration of plasma reactor [40]. (1) reactor shell; (2) fixing devices; (3) ground electrode; (4) high-voltage electrode; (5) dielectric.

3.2. Microwave Discharge

The microwave discharge method is an advanced technology that utilizes high-frequency electromagnetic waves (typically 2.45 GHz) to generate plasma, facilitating chemical reactions and showing great potential in hydrogen production. This method efficiently breaks chemical bonds in target substances through the action of high-energy electrons and active free radicals, and is applicable to various gaseous and liquid fuels, including methane [43], methanol [44], and ethanol [45]. In recent years, extensive research into the reforming and cracking of various fuels using microwave discharge has occurred. For example, in the composite reforming of methane, reactions involving steam and carbon dioxide can achieve efficient hydrogen production, with an energy yield of up to 42.9 g(H₂)/kWh [44]. This technology can also be applied in liquid-phase systems, where direct discharge reactions in ethanol solutions achieve a hydrogen production rate of 72.48 g/h and energy efficiency of 48.32 g/kWh [46]. The study in [47] demonstrated that microwave discharge provides higher hydrogen yields in liquid-phase reactions compared to traditional methods, with lower energy consumption. These advantages make microwave discharge a promising method for hydrogen production. Additionally, a recent study [48] explored the ultra-high growth rate of boron-doped diamond films using in-liquid microwave plasma CVD, showcasing the versatility of microwave plasma technology in enhancing material properties alongside its application in hydrogen production. These advantages make microwave discharge a promising method for hydrogen production.

The microwave discharge method is characterized by rapid startup and high efficiency, offering advantages in various practical applications. Figure 6 shows a microwave plasma hydrogen production system, which includes a microwave discharge system, measurement system, gas supply and flow control system, gas analysis system, and Optical Emission Spectroscopy (OES) system. Unlike conventional thermal catalytic techniques, this method does not require complex preheating or catalysts, significantly reducing startup time while efficiently generating high-energy electrons at ambient temperature [49]. Furthermore, its high selectivity and reaction rate benefit from the rich variety of active species and high energy density within the plasma, significantly enhancing hydrogen generation efficiency. Additionally, since microwave discharge is applicable to both liquid and gaseous fuels, it eliminates the need for an additional carrier gas system, thus reducing equipment complexity. The liquid-phase discharge also prevents catalyst poisoning and carbon deposition, effectively lowering maintenance costs over long-term operation [50].

However, the microwave discharge method also faces certain limitations in practical applications. One of the primary challenges is energy efficiency, particularly when processing high-flow fuels, where the energy efficiency often fails to meet industrial requirements. Moreover, the design and optimization of microwave discharge reactors require consideration of multiple variables, including microwave power, gas flow, and reactor structure, which increases experimental optimization complexity and initial investment costs [51]. To address these issues, current research primarily focuses on improving energy utilization efficiency by optimizing coupling structures and reaction conditions. For example, Chehade [20] and colleagues suggested that improving the matching between microwave radiation and the reaction medium in the reactor, or incorporating heterogeneous catalysts, could further enhance hydrogen production rates.

Additionally, there remains some debate in the academic community regarding the mechanisms of the microwave discharge method, especially concerning the role of water molecules in liquid-phase discharge. Research by Zhu [50] suggests that hydroxyl free radicals in water can effectively suppress the formation of certain by-products and significantly enhance hydrogen yield, while research by Wang [52] offers alternative hypotheses regarding their specific role. Therefore, one of the future research focuses will be to further

elucidate the reaction pathways in liquid-phase microwave discharge, clarifying the precise roles of each active species to enable more accurate control of the reaction process.

The microwave discharge method, with its rapid, efficient, and flexible characteristics, has become an important research direction in the field of hydrogen production. In future developments, this technology is expected to become a hydrogen production method with industrial scalability, driven by reactor design optimization, improved energy efficiency, and the expansion of application scenarios. Particularly in real-time hydrogen production for mobile devices (such as vehicles and ships), the microwave discharge method, with its advantages of compact device size and fast response, demonstrates unique potential.

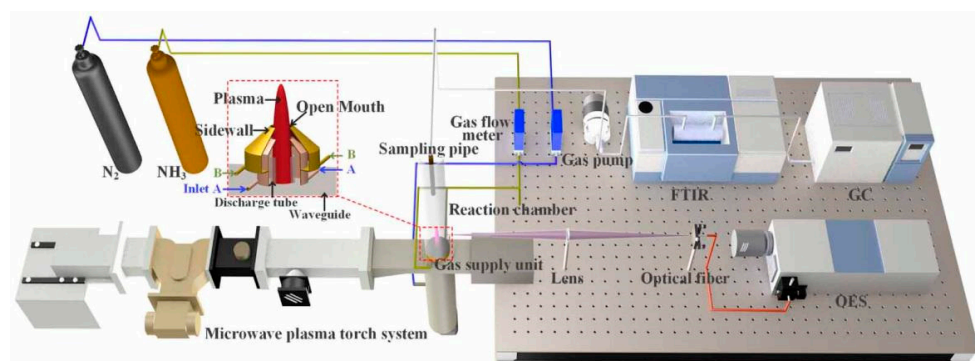


Figure 6. Schematic of experimental setup [51].

3.3. Dielectric Barrier Discharge (DBD)

The dielectric barrier discharge (DBD) plasma reactor consists of two parallel electrodes, with at least one electrode surface covered by a dielectric layer. The role of the dielectric is to limit the direct transmission of charges while ensuring a uniform electric field distribution within the discharge region, effectively preventing spark formation. During the DBD discharge process, the voltage difference between the electrodes generates reactive species, composed of electrons, ions, and neutral particles. The induced electric field accelerates the electrons toward the positive electrode, causing them to collide with gas molecules, leading to ionization, excitation, and dissociation. Ionizing collisions not only generate new electrons that rapidly accelerate toward the negative electrode but also, through the secondary electron emission phenomenon, further trigger new ionization collisions. The secondary ionization phenomenon helps maintain the plasma discharge state continuously [53].

Based on this, Hu et al. [54] conducted a study on ethanol decomposition for hydrogen production using DBD technology. The results showed that the main products of ethanol decomposition included H_2 , CO , CH_4 , CO_2 , C_2H_4 , and C_2H_6 . Further experiments revealed that the efficiency of ethanol reforming was significantly improved when glass beads were filled in the gap between the inner and outer electrodes. This was because the inclusion of glass beads enhanced the electric field effect while increasing the residence time of ethanol in the discharge region, thereby promoting the reaction. Under optimized experimental conditions, when the feed flow rate was 5 mL/min and the diameter of the glass beads was 2 mm, the highest hydrogen production efficiency was achieved, with an ethanol conversion rate of up to 45%.

Meanwhile, Wang et al. [55] also conducted similar research on ethanol decomposition for hydrogen production using dielectric barrier discharge (DBD) technology. The study demonstrated that both the ethanol conversion rate and hydrogen selectivity were significantly influenced by the gasification temperature, and that as the feed flow rate increased, both the conversion rate and selectivity decreased. The authors explained that, under

constant power conditions, an increase in ethanol flow resulted in a reduction in the decomposition energy available to each ethanol molecule, preventing effective decomposition. Additionally, the study found that hydrogen selectivity increased with the water-to-ethanol molecule ratio, while the addition of oxygen helped improve ethanol conversion. Under optimal experimental conditions, the ethanol conversion rate reached 88.4%.

Furthermore, Sato [56] explored the methanol decomposition process for hydrogen production using DBD technology from a microscopic perspective. The results indicated that reactive species such as $O\cdot$, $OH\cdot$, and $H\cdot$ were generated during the reaction, and these species played a key role in promoting methanol decomposition. Further research confirmed that $OH\cdot$ was the most important active species in the methanol decomposition process.

Taghvaei et al. used a nanosecond pulse DBD plasma reactor for hydrogen production from heavy naphtha. The experimental setup, shown in Figure 7, studied the effects of carrier gas flow, feed flow rate, and discharge gap on the gas production. The results indicated that the main products of DBD plasma decomposition of heavy naphtha were H_2 and hydrocarbons with carbon numbers ranging from C_1 to C_3 , with the product distribution ordered as $C_2 > C_1 \gg C_3 > C_4$. Additionally, they found that lower carrier gas and feed flow rates led to higher energy efficiency. In their experiment, under conditions where the inner electrode diameter was 2.68 mm and the carrier and feed flow rates were 100 mL/min and 1 mL/min, respectively, the hydrogen production rate was 9.26 mL/min, and the energy efficiency of the cracking process reached 159.29 L/kWh.

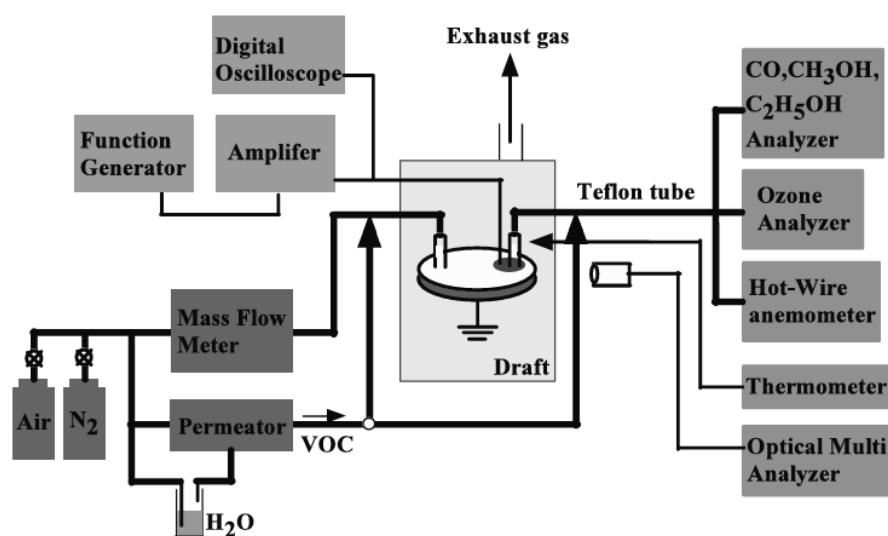


Figure 7. Schematic of experimental apparatus [56].

3.4. Radio Frequency Discharge

Radio Frequency (RF) plasma is a promising technology for hydrogen production due to its efficiency, low environmental impact, and flexibility in operating conditions. RF plasma is generated by applying high-frequency electromagnetic fields, typically at 13.56 MHz, to ionize gases such as methane or water vapor. This process creates reactive species like high-energy electrons and free radicals that can break molecular bonds, enabling the dissociation of methane (CH_4) and water (H_2O) into hydrogen gas (H_2) and other by-products.

In the case of methane decomposition, RF plasma can efficiently crack methane molecules into hydrogen and solid carbon. Studies have shown that at RF power levels ranging from 500 to 1300 W, hydrogen yields can reach up to 95–100%, with methane conversion efficiencies exceeding 99% [57]. This method presents a significant advantage

over traditional steam methane reforming (SMR), as it avoids carbon dioxide emissions and operates at lower temperatures.

For water vapor dissociation, RF plasma offers a more energy-efficient alternative to electrolysis. The high-energy electrons in RF plasma break the O-H bonds in water molecules, producing hydrogen and oxygen. Hydrogen purity of up to 70% has been achieved under optimal conditions [58]. The ability to control RF power and reactor conditions allows for precise optimization of the hydrogen production rate, making RF plasma a versatile option for both small-scale and industrial hydrogen production. Figure 8 shows the process of hydrogen production from subsea hydrate fields using RF plasma.

Despite its potential, RF plasma faces challenges such as high energy consumption per kilogram of hydrogen produced and the need for further optimization in reactor design for scalability. However, as research progresses, RF plasma is expected to become a more cost-effective and sustainable method for hydrogen production, contributing to a greener hydrogen economy.

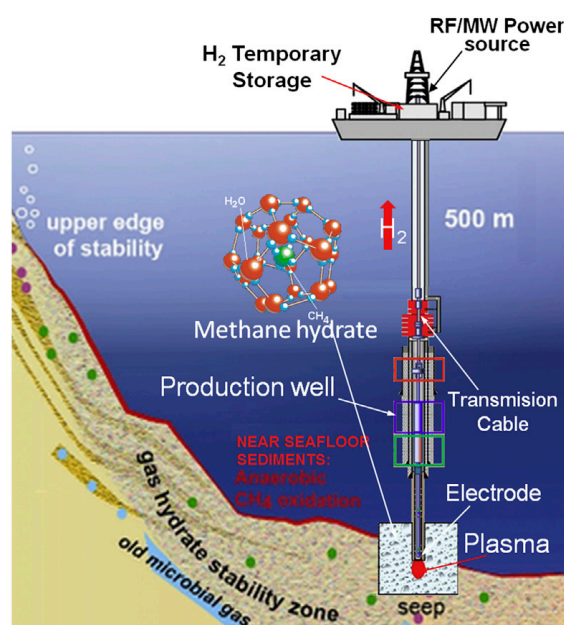


Figure 8. The process of hydrogen production from underwater hydrate oil fields using the liquid plasma method [36].

3.5. Comparison of Hydrogen Production and Energy Efficiency

This section presents a comprehensive comparison of hydrogen production rates and energy efficiency among various plasma discharge technologies, including arc discharge (AD), microwave discharge, dielectric barrier discharge (DBD), and radio frequency (RF) plasma, as summarized in the table below. The data reveal significant differences in hydrogen production yield and energy efficiency depending on the plasma discharge method and the feedstock used.

The table illustrates that microwave discharge typically achieves the highest hydrogen production rates, particularly in methane and ammonia decomposition, with yields of 74.0% and 54.4%, respectively [45–51]. Arc discharge, on the other hand, shows more variability in hydrogen yield depending on the feedstock, with a maximum of 48.6% for methane decomposition [39]. DBD excels in ammonia decomposition, achieving an exceptionally high hydrogen yield of 99.9% [59], whereas RF plasma shows relatively lower hydrogen production rates, with a maximum of 64.2% for methane and steam decomposition [33].

In terms of energy efficiency, arc discharge outperforms the other methods, particularly in methane decomposition, where it achieves an energy efficiency of 136.6 L/(kW·h), highlighting its potential for efficient hydrogen production. RF plasma, however, shows significantly lower energy efficiency at 6.72 L/(kW·h) [36]. DBD performs well in ammonia decomposition with an energy efficiency of 430 L/(kW·h) [59], underscoring its potential for low-energy hydrogen production.

From an industrial perspective, arc discharge is favored for large-scale applications due to its maturity and adaptability to a wide range of feedstocks, though fluctuations in yield and efficiency require process optimization. Microwave discharge is effective for liquid-phase reactions but is hindered by high equipment costs and energy consumption, making it more suitable for distributed hydrogen production or specific applications. DBD holds industrial promise but faces challenges in catalyst stability and reactor design. RF plasma, with its relatively low energy efficiency, is less commonly used in industrial settings and may require integration with catalytic technologies to enhance its competitiveness in the future.

In summary, DBD and microwave discharge offer excellent hydrogen production rates and energy efficiency under specific conditions; however, their industrial implementation requires optimization of cost and stability. Arc discharge remains the predominant choice for large-scale applications, while RF plasma is still in the experimental stage. It is important to note that variations in experimental conditions and the lack of certain data may affect the generalizability of these findings, and actual applications should be adjusted according to specific process parameters (Table 1).

Table 1. Comparison of different discharge modes.

Hydrogen Technology	Items	Feedstock	H ₂ Yield (%)	Energy Yield (L·(kW·h) ^{−1})	References
Plasma generation techniques	Arc discharge	C ₇ H ₈	48.6	60.2	[39]
	Arc discharge	NH ₃	34.8	1080.0	[42]
	Arc discharge	C ₂ H ₆ O	40.9	-	[38]
	Arc discharge	C ₇ H ₁₆	34–53	77–89	[60]
	Microwave discharge	C ₂ H ₆ O	58.1	23.97	[50]
	Microwave discharge	CH ₄	74.0	136.6	[45]
	Microwave discharge	CH ₄	9.5	21.28	[44]
	Microwave discharge	NH ₃	54.4	274	[51]
	Dielectric barrier discharge	NH ₃	15.1	48.6	[61]
	Dielectric barrier discharge	NH ₃	99.9	430	[59]
	Dielectric barrier discharge	CH ₄ and H ₂ O	80	-	[62]
	Radio frequency discharge	CH ₄ and H ₂ O	64.2	6.72	[36]

4. Applications for Plasma-Based Hydrogen Production

4.1. Water Splitting for Hydrogen Production

The raw material for the electrolysis-based hydrogen production method is water, and the water resources on Earth are abundantly distributed. Using this method for hydrogen production does not result in the depletion of raw materials or the dilemma of non-renewable resources. Currently, industrial hydrogen production via water electrolysis constitutes 4% of the total hydrogen production. The primary energy consumption in the electrolysis hydrogen production process is electricity, and the electricity cost for producing

1 m³ of H₂ represents approximately 80% of the total electrolysis hydrogen production cost [63].

Currently, the primary methods for hydrogen production through electrolysis include alkaline water electrolysis (AWE), proton exchange membrane water electrolysis (PEM), alkaline anion exchange membrane water electrolysis (AEM), solid oxide electrolysis (SOEC), and plasma water electrolysis (PWE). Table 2 below compares and summarizes these technologies [64–71] and Table 3 presents the advantages and disadvantages of these technologies.

Table 2. Comparison of energy consumption for different hydrogen production technologies.

Hydrogen Technology	Items	Energy Consumption (\$/GJ)	References
Plasma-assisted hydrogen production methods	Plasma-assisted steam cracking	30.75	[15]
	Plasma-assisted hydrocarbon cracking	0.83	[72]
Plasma generation techniques	Arc discharge	12.81	[73]
	Microwave discharge	2.8–5.1	[57]
	Dielectric barrier discharge	7.53–11.39	[21]
	Radio frequency discharge	0.88–1.38	[57]

Table 3. Comparison of electrolyzed water to hydrogen technologies.

Hydrogen Technology	Working Temperature/(°C)	Advantages	Disadvantages	Industrialization
AWE	30~80	low cost quick response;	low current density	industrialize
PEMWE	55~65	low power consumption;	high water quality requirements	special applications
AEMWE	40~60	low cost	poor thermal stability	laboratory phase
SOEC	700~900	high electrolysis efficiency	rapid material degradation; short operating life; high manufacturing costs	laboratory phase
PWE		energy consumption	high requirements for equipment	early stage of commercialization

Plasma electrolytic water hydrogen production technology is being progressively industrialized. It is highly safe and efficient, capable of adapting to fluctuations in renewable energy power. Plasma electrolytic water hydrogen production may be conducted at various temperatures, depending on the experimental conditions and equipment design. However, compared to the traditional high-temperature steam cracking method, it can be performed at a lower temperature, thereby enhancing energy conversion efficiency. In plasma electrolysis, the metal electrode responsible for hydrogen production is no longer immersed in the water, as is the case in conventional water electrolysis. Instead, the electrode interacts with the water through the plasma situated between the top of the electrode and the water surface to produce hydrogen [74]. Under specific experimental conditions, the hydrogen production efficiency of plasma is significantly higher than that of traditional electrolysis. For instance, Yulianto and colleagues found that the hydrogen production efficiency of plasma electrolysis at 120 V is significantly higher than that of conventional electrolysis at 300 V, with a difference of up to 80 times [75,76].

Although plasma electrolytic technology has demonstrated its advantages in hydrogen production, scholars are also investigating the integration of plasma with other hydrogen production methods to further reduce costs and improve efficiency [16]. Beyond its use in electrolysis, plasma can also be applied in other hydrogen production methods from water, such as pyrolysis, photolysis, and photo/electrocatalysis [77].

For instance, Wang et al. [78] prepared high-performance transition metal-based electrocatalysts via plasma treatment, which exhibited excellent electrocatalytic activity and stability, with a hydrogen production rate significantly higher than that of most other transition metal-based electrocatalysts. Plasma can also be applied in photocatalysis to emit OH radicals [79] through water (with methanol as an additive) or to modify surfaces and create active sites. Ding et al. [80] employed low-temperature plasma technology to synthesize the Pt/g-C₃N₄ photocatalyst, which significantly enhanced its photocatalytic hydrogen production efficiency. The photocatalytic hydrogen production efficiency of Pt/g-C₃N₄ was 63.2 and 4.6 times higher than that of the original g-C₃N₄ and the Pt/g-C₃N₄ composite material synthesized using the traditional photo deposition method.

Further research into this technology will provide additional complementary and enhanced functionalities for plasma-assisted water electrolysis for hydrogen production.

Although the use of catalysts can lead to a significant increase in hydrogen production, there is ongoing debate regarding their commercially viable potential in supporting plasma hydrogen production [81]. Electrolytic water enables efficient operation in small-scale hydrogen production, which can be integrated with renewable energy sources for energy storage [82,83]. Moreover, plasma electrolytic water hydrogen production holds the potential for integration into fuel cell systems and modular units in the future, thereby enhancing small-scale power systems for distributed power generation.

4.2. Plasma Cracking of Natural Gas

Plasma cracking of natural gas, particularly methane, for hydrogen production not only enables efficient hydrogen generation but also reduces environmental pollution by addressing by-products such as carbon black. As a result, it is regarded as a hydrogen production technology with enormous potential. In recent years, much research has focused on optimizing the plasma cracking process to improve methane conversion rates, hydrogen yield, and system energy efficiency.

As early as 2012, Abánades et al. [84] conducted an in-depth review of the technological challenges facing the industrial development of methane cracking for hydrogen production. For direct thermal cracking applications, the primary issue is preventing carbon particle formation, which can lead to equipment blockages. To address this, the researchers suggested optimizing system performance through the proper design of reactors and the development of suitable gas turbulence or porous media. In catalytic thermal cracking, the economic feasibility of the catalyst remains a key challenge, limiting its widespread industrial application. Subsequently, Barni et al. [85] successfully achieved a methane conversion rate of 60% using the nanosecond pulsed dielectric barrier discharge (DBD) technology, with a hydrogen yield of approximately 25% and a corresponding hydrogen yield of 450–600 kJ/g. The study also conducted experiments on highly diluted argon/methane mixtures using a sinusoidal DBD reactor, showing that argon dilution significantly improved the plasma methane reforming performance. Additionally, numerical simulations were used to verify the plasma chemical kinetics, providing theoretical support for further optimization of the technology.

Nickel-based catalysts are widely utilized in various reforming processes due to their excellent activity, selectivity, and relatively low cost compared to other metals such as platinum or rhodium [86]. Nickel catalysts are particularly effective in the steam reforming

of methane (SMR) and dry reforming of methane (DRM), where they facilitate the breaking of C-H bonds in methane molecules [87]. In these processes, nickel catalysts are typically supported on materials such as alumina (Al_2O_3), magnesium aluminate (MgAl_2O_4), or ceria (CeO_2) to enhance their thermal stability and resistance to carbon deposition.

The use of nickel catalysts in plasma-assisted methane cracking has shown promising results, where they not only improve methane conversion rates but also help in suppressing undesirable by-products like carbon deposition, which is a significant challenge in traditional methane cracking processes [88]. The combination of plasma with nickel catalysts in methane cracking reactions offers an innovative approach to achieving high hydrogen yields, increasing the overall energy efficiency, and reducing the formation of coke and other carbonaceous materials.

Putra et al. [89], by comparing with methane steam reforming (SMR), found that the methane cracking reaction (MCR) is the primary route for converting methane into hydrogen. At an input power of 150 W, they achieved a 40% methane conversion rate and a hydrogen content of 55%, resulting in a hydrogen yield of 41%. These results indicate that plasma methane cracking holds significant potential for hydrogen production. Taghvaei et al. [72] further investigated the influence of hydrocarbon type on plasma cracking performance. Using a nanosecond pulsed DBD reactor, they found that the performance of the reactor significantly improved with an increase in the carbon number of the hydrocarbon feed. The cracking energy efficiency and hydrogen production rate reached 23.8–121.1 L/kWh and 17.04–34.05 mL/min, respectively, indicating that plasma cracking exhibits high energy efficiency and gas production capacity under various hydrocarbon feed types. Khalifeh et al. [90] studied the process of combining nanosecond pulsed plasma with Pt-Re catalysts for methane cracking to produce high-purity hydrogen. In their experiments, methane and argon flow rates were 20 mL/min and 50 mL/min, respectively. The results showed that under conditions near 9 W of power, the plasma system achieved the production of ultra-pure hydrogen with 100% methane conversion. At a power consumption of about 4 W, the standalone plasma system also achieved 26.08% energy efficiency. The study also demonstrated that using lower argon flow rates and power below 10 W not only resulted in extremely high energy efficiency and methane conversion rates but also effectively suppressed the generation of higher hydrocarbon by-products and carbon deposition on the electrode surfaces. Khoja et al. [91], in their research on plasma-catalyzed methane cracking for hydrogen production, used a Ni/ MgAl_2O_4 nano-catalyst and found that the plasma-catalyzed process significantly increased methane conversion (reaching 80%), improving hydrogen selectivity to 75%. In contrast, the standalone plasma cracking system achieved only a 62% methane conversion rate. This study demonstrated that plasma-catalytic integration significantly enhances cracking efficiency, improving hydrogen yield while suppressing by-product formation.

Furthermore, Gao et al. [92] developed a novel Ni-Ce/ZSM-5 catalyst for the conversion of methane and carbon dioxide. The study showed that using this catalyst, methane conversion rates could reach 99%, and carbon dioxide conversion rates could reach 94%. The catalyst demonstrated excellent stability during long-term operation, maintaining a methane conversion rate above 95% and a carbon dioxide conversion rate above 85% over 40 h. The H_2/CO ratio in the syngas was close to 1, further confirming the catalyst's potential in plasma cracking applications. Wang et al. [93] investigated the performance of methane and carbon dioxide reforming reactions by analyzing electrical parameters. They found that increasing the pulse frequency significantly improved the conversion rates of both methane and carbon dioxide. When the pulse peak width was 150 ns, the conversion rates of methane and carbon dioxide reached their maximum values. Under conditions of a 10 kHz pulse repetition frequency and a discharge power of 55.7 W, the maximum

conversion rates of methane and carbon dioxide were 39.6% and 22.9%, respectively, while the total energy conversion efficiency of the syngas was 5.0% and 7.1%, respectively.

Finally, Sun et al. [94] studied the effects of plasma-generated active species on the surface chemistry and coke formation in methane and carbon dioxide reforming reactions under reduced pressure (8–40 kPa) conditions through steady-state experiments and a global model. The experimental results showed that the coupling of a Ni/SiO₂ catalyst with plasma was significantly more effective than an unfilled DBD plasma system. Under conditions of 8 kV and 473 K, the total conversion rate of the catalytic system reached 16%, with corresponding CO and H₂ yields of 15% and 12%, respectively.

In summary, plasma cracking of natural gas (particularly methane) for hydrogen production has demonstrated broad application prospects in enhancing hydrogen yield, conversion efficiency, and reducing by-product formation. By optimizing the plasma cracking process, exploring innovative coupling strategies between catalysts and plasma, and improving system energy efficiency, this technology holds promise for providing more efficient and environmentally friendly solutions for future hydrogen production (Figure 9).

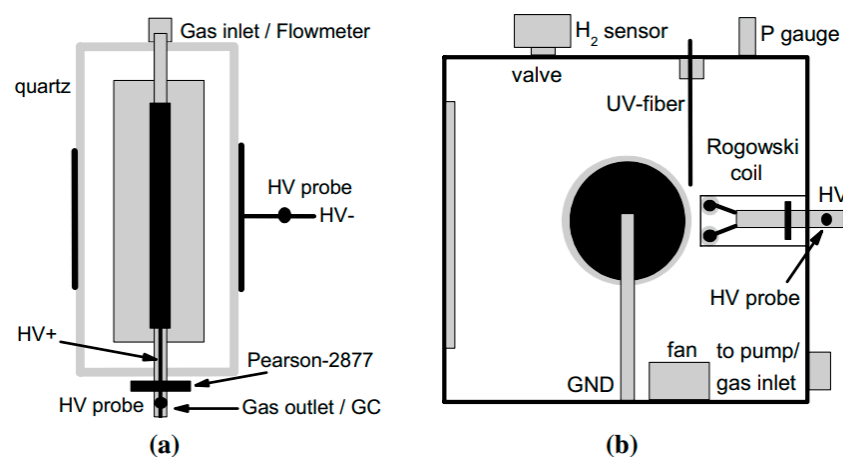


Figure 9. Schematic view of the experimental setup and diagnostics of the plasma reactors: Milano CNR Laboratory (a), Milano-Bicocca Laboratory (b) [85].

4.3. Waste Gas Resource Utilization

Plasma technology has demonstrated immense potential in resource utilization, particularly in the treatment of industrial exhaust gases and municipal waste. The high-energy plasma process effectively transforms harmful substances in waste gases, including carbon dioxide and nitrogen oxides, into usable energy or valuable chemicals. For instance, plasma pyrolysis technology can convert organic waste into syngas, hydrogen, methane, and other clean energy sources, while plasma gasification can convert municipal waste into combustible gases and liquid fuels through high-temperature processing. In this manner, not only is waste harmlessly disposed of, but its energy resources are effectively recovered, offering innovative solutions for sustainable development.

In the plasma gasification process, materials are gasified in an oxygen-limited environment, decomposing the waste into fundamental molecular components. Unlike incinerators, it does not burn waste. Electricity is supplied to a torch containing two electrodes, generating an electric arc. The continuous electric current within the plasma generates an energy-intensive field, powerful enough to break down the waste into fundamental elements. The byproduct is a glass-like material, which can serve as a raw material for high-strength asphalt or household tiles, in addition to syngas. Syngas, a mixture of hydrogen and carbon monoxide, can be converted into hydrogen, natural gas, or ethanol-based fuels. The produced syngas is directed to a cooling system, where it generates steam. The

steam drives turbines to generate electricity, part of which powers converters, while the remainder can be used for plant heating, power generation, or sold to the grid. Metals are melted, and inorganic materials, including silica, soil, concrete, glass, and gravel, are vitrified, flowing out from the reactor bottom. There is no need to landfill tar, furans, or ash [95–98].

Gasification is a straightforward and commercially established technology. It involves converting a variety of feedstocks into clean syngas through reactions with oxygen and steam. The reaction occurs spontaneously at elevated temperatures and pressures under reducing conditions, consuming half the oxygen required for complete combustion. After cooling and purifying the syngas, it can be utilized for one or more of the following applications: Syngas can be used to produce chemicals and gaseous fuels, combusted in commercial boilers to generate steam, or employed in heat transfer processes and internal combustion engines to generate electricity. Additionally, combined-cycle systems can enable cogeneration of heat and power [97,99,100].

The gasification process can occur in a closed plasma reactor, which is a sealed stainless-steel vessel filled with ambient air. A 650-volt current flows between two electrodes, ionizing the air and generating plasma. The plasma reactor does not differentiate between any types of waste. The only variable is the energy required to process the waste. Therefore, waste does not need to be classified; any type of waste, except nuclear waste, can be processed [101].

Please see Figure 10 below for a flow chart of the process of waste gas and waste resourcing through plasma technology.

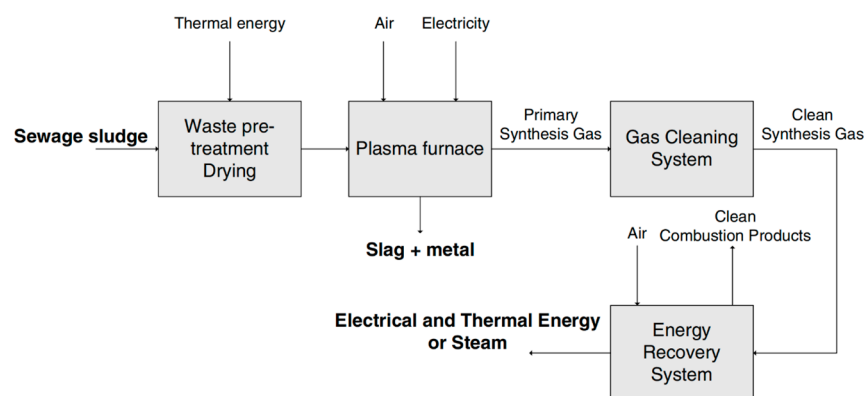


Figure 10. Block diagram of plasma gasification of sewage sludge [95].

5. Conclusions

As global demand for clean energy rises, hydrogen is becoming a key component of the energy transition. Traditional hydrogen production methods, relying on fossil fuels, are inefficient and environmentally harmful. Plasma-assisted hydrogen production, as a new and promising technology, offers high efficiency, environmental benefits, and flexibility. This review discusses the principles, technologies, advancements, and challenges of plasma-assisted hydrogen production, highlighting its potential advantages and future prospects.

Plasma technologies generate high-energy electrons and ions to decompose water vapor or hydrocarbons into hydrogen. Methods such as steam reforming, hydrocarbon cracking, arc discharge, and microwave discharge can optimize hydrogen yield and energy efficiency under various conditions. Although laboratory results are promising, scaling these methods to industrial levels faces challenges such as energy efficiency, reactor stability, and cost.

Despite its early stage of commercialization, plasma-assisted hydrogen production has the potential to become a mainstream hydrogen production method, supporting global low-carbon strategies and reducing greenhouse gas emissions.

Looking ahead, plasma-assisted hydrogen production is expected to play a central role in energy systems, particularly with the increasing integration of renewable energy sources such as solar and wind power. Combining plasma technologies with renewable energy can enable efficient, low-carbon hydrogen production, contributing to achieving carbon neutrality goals. Additionally, the development of hybrid plasma-catalytic systems, optimized reactor designs, and advanced materials could significantly improve energy efficiency, lower costs, and enhance the scalability of plasma-assisted hydrogen production. These advancements will be key to making plasma-assisted hydrogen production a widely adopted technology in the future.

In the future, plasma-assisted hydrogen production could play a key role in energy systems, especially when integrated with renewable energy and distributed generation. With further development, it could help achieve carbon neutrality and sustainable development goals.

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